MODULE 09: LC CIRCUIT & MAXWELL EQUATIONS

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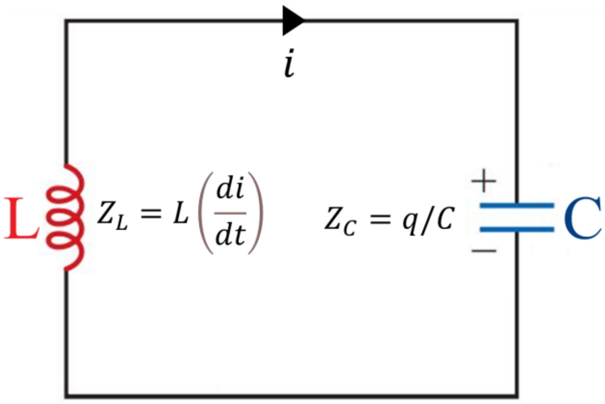
LECTURE 20

OUTLINE:
- LC circuit

LC CIRCUIT

An LC circuit, also known as a resonant circuit, tank circuit, or tuned circuit, is an electric circuit consists of two passive components: an inductor (L) and a capacitor (C) connected together.

These components are connected in parallel or series, and the circuit can exhibit interesting electrical behavior based on the values of the inductance and capacitance. LC circuits are commonly used in various applications, including in radio receivers, signal filtering, and oscillators.



$$L\left(\frac{di}{dt}\right) + \frac{q}{C} = 0$$

$$L\left(\frac{d^2q}{dt^2}\right) + \frac{q}{C} = 0$$

$$q = Q_m \cos(\omega t + \phi)$$

The energy stored in one location can be transferred to another location. In an oscillating LC circuit, energy is shuttled periodically between the electric field of the capacitor and the magnetic field of the inductor.

The energy stored in the electric field of the capacitor at any time, knowing q is the charge on the capacitor at that time, is given by,

$$U_E = \frac{1}{2}\frac{q^2}{C}$$

The energy stored in the magnetic field of the inductor at any time, knowing i be the current through the inductor at that time, is given by,

$$U_B = \frac{1}{2}i^2 L$$

LC OSCILLATIONS

LC oscillations, also known as LC tank circuit oscillations, refer to the natural oscillatory behavior exhibited by an LC circuit when excited by an initial energy source and then left to resonate freely.

- In the LC circuit, charge, current, and potential difference vary sinusoidally (with period T and angular frequency ω). The resulting oscillations of the electric field of the capacitor and the magnetic field of the inductor are said to be ***electromagnetic oscillations***.
- LC oscillations have various applications, including in the generation of precise frequencies in electronic oscillators, as well as in radio receivers for tuning to specific frequencies. They are also used in many signal processing and filtering applications.
- The succeeding stages of the oscillations of a simple *LC* circuit are presented in *Fig 1*.

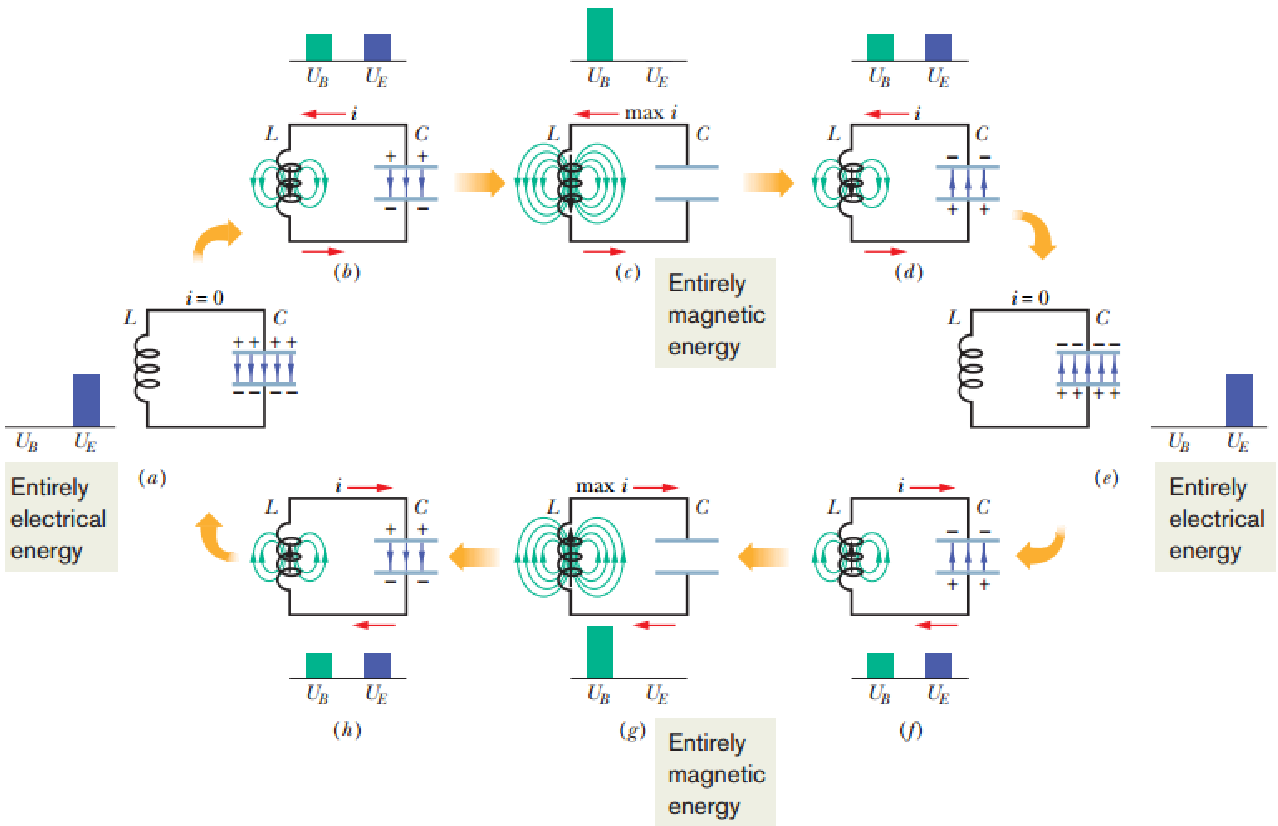


Fig 1. *Eight stages in a single cycle of oscillation of a resistance less LC circuit. The bar graphs by each figure show the stored magnetic and electrical energies. The magnetic field lines of the inductor and the electric field lines of the capacitor are shown. (a) Capacitor with maximum charge, no current. (b) Capacitor discharging, current increasing. (c) Capacitor fully discharged, current maximum. (d) Capacitor charging but with polarity opposite that in (a), current decreasing. (e) Capacitor with maximum charge having polarity opposite that in (a), no current. (f) Capacitor discharging, current increasing with direction opposite that in (b). (g) Capacitor fully discharged, current maximum. (h) Capacitor charging, current decreasing*

Initiation of a LC oscillations, requires initial charging of the capacitor (C) with some energy. This can be achieved, for example, by connecting a voltage source across the LC circuit. The capacitor stores energy in its electric field, and the inductor stores energy in its magnetic field.

Let us consider that initially the charge (q) on the capacitor is at its maximum value Q and that the current (i) through the inductor is zero, as shown in $Fig. 1a$.

- The bar graphs for energy indicate that at this instant, with zero current through the inductor and maximum charge on the capacitor, the energy U_B of the magnetic field is zero and the energy U_E of the electric field is a maximum.
- As the circuit oscillates, energy shifts back and forth from one type of stored energy to the other, but the ***total amount is conserved***.

The capacitor then starts to discharge through the inductor. The positive charge carriers move counterclockwise, as shown in $Fig.\,1b$.

- Thus, a current i, given by $\frac{dq}{dt}$ and pointing down in the inductor, is established.
- As the charge in the capacitor decreases, the energy stored in the electric field within the capacitor also decreases.
- This energy is transferred to the magnetic field that appears around the inductor because of the current i that is building up there.
- Thus, the electric field decreases, and the magnetic field builds up as energy is transferred from the electric field to the magnetic field.

Eventually, the capacitor loses all its charge (Fig. *c*) and thus also loses its electric field and the energy stored in that field.

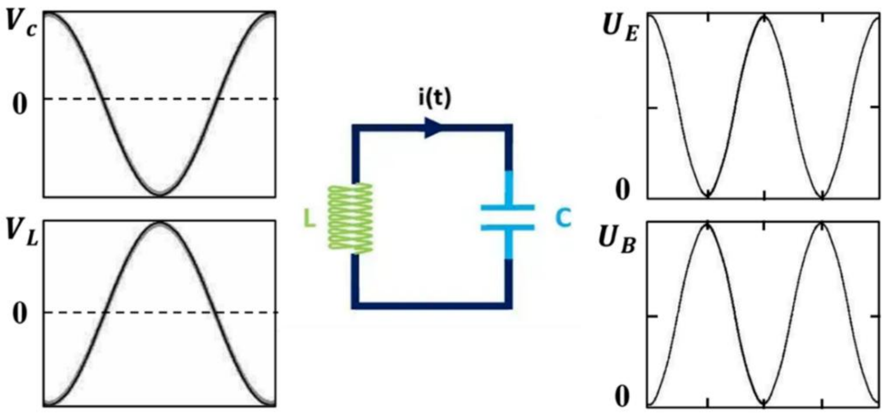
- The electric energy has then fully been transferred to the magnetic field of the inductor.
- The magnetic field is then at its maximum magnitude, and the current through the inductor is then at its maximum value I.

Although the charge on the capacitor is zero at this stage, the counterclockwise current must continue because the inductor does not allow it to change suddenly to zero.

- The current continues to transfer positive charge from the top plate to the bottom plate through the circuit ($Fig.\,1d$).
- Energy now flows from the inductor back to the capacitor as the electric field within the capacitor builds up again. The current gradually decreases during this energy transfer.
- When, eventually, the energy has been transferred completely back to the capacitor ($Fig.\,1e$), the current has decreased to zero (momentarily).
- The situation of $Fig.1e$ is like the initial situation, except that the capacitor is now charged oppositely.

The capacitor then starts to discharge again but now with current in a clockwise ($Fig.1f$). The clockwise current builds to a maximum ($Fig.\,1g$) and then decreases ($Fig.1h$), until the circuit eventually returns to its initial situation (Fig. *a*).

- The process then repeats at some frequency f and thus at an angular frequency $\omega = 2\pi f$.
- In the ideal LC circuit with no resistance, there are no energy transfers other than that between the electric field of the capacitor and the magnetic field of the inductor.
- Because of the conservation of energy, the oscillations continue indefinitely.
- The oscillations need not begin with the energy all in the electric field; the initial situation could be any other stage of oscillation.



The charge q on the capacitor as a function of time can be expressed with respect to the time-varying potential difference (V_C) across the capacitor C as follows,

$$V_C = \frac{q}{C} \qquad \Rightarrow q = CV_C$$

Similarly, the current can be measured across a small resistance R connecting in series with the capacitor and inductor. The time-varying potential difference V_R can also be measured across the resistor. The V_R is proportional to i through the relation,

$$V_R = iR \qquad \Rightarrow i = \frac{V_R}{R}$$

We assume here that R is so small that its effect on the behavior of the circuit is negligible. The variations in time of V_C and V_R, and thus of q and i. All four quantities vary sinusoidally.

- In an actual LC circuit, the oscillations will not continue indefinitely because there is always some resistance present that will drain energy from the electric and magnetic fields and dissipate it as thermal energy (the circuit may become warmer). The oscillations, once start, will die away.

CHECKPOINT: A charged capacitor and an inductor are connected in series at time $t = 0$. In terms of the period T of the resulting oscillations, determine how much later the following reach their maximum value: (a) the charge on the capacitor; (b) the voltage across the capacitor, with its original polarity; (c) the energy stored in the electric field; and (d) the current.

ANSWER: $(a)\ T/2;\ (b)\ T;\ (c)\ T/2;\ (d)\ T/4$

LC OSCILLATOR

An LC oscillator is an electronic oscillator circuit that uses a combination of inductance (L) and capacitance (C) components to generate and sustain oscillations at a specific frequency. LC oscillators are a type of harmonic oscillator, and they are widely used in various electronic devices and systems.

For a resistance-less LC circuit, the total energy U at any instant in an oscillating LC circuit is,

$$U = U_B + U_E = \frac{1}{2}i^2L + \frac{q^2}{2C}$$

Here, U_B is the energy stored in the magnetic field of the inductor and U_E is the energy stored in the electric field of the capacitor. Since, the resistance of the circuit is assumed to be zero, no energy is transferred to thermal energy and U remains constant with time. Therefore, $\frac{dU}{dt}$ must be zero. Thus,

$$\frac{dU}{dt} = \frac{d}{dt}\left(\frac{1}{2}i^2L + \frac{q^2}{2C}\right) = iL\frac{di}{dt} + \frac{q}{C}\frac{dq}{dt} = 0$$

However, $i = \frac{dq}{dt}$ and $\frac{di}{dt} = \frac{d^2q}{dt^2}$. With these substitutions, equation becomes,

$$L\frac{d^2q}{dt^2} + \frac{q}{C} = 0$$

This is the *differential equation* that describes the oscillations of a resistance-less LC circuit.

Charge and Current Oscillations:

In LC oscillations, both charge and current vary periodically as the energy stored in the electric field of the capacitor (C) and the magnetic field of the inductor (L) continually transfer back and forth.

$$\frac{d^2q}{dt^2} + \frac{1}{LC}q = 0$$

The solution of the differential equations gives,

$$q = Q\cos(\omega t + \phi)$$

where Q is the amplitude of the charge variations, ω is the angular frequency of the electromagnetic oscillations, and ϕ is the phase constant.

Taking the first derivative with respect to time gives us the current,

$$i = \frac{dq}{dt} = -\omega Q\sin(\omega t + \phi) = -I\sin(\omega t + \phi)$$

Here, $I = \omega Q$ is the amplitude *is* of this sinusoidally varying current.

Angular Frequencies:

The angular frequency of an LC circuit is a measure of how fast the circuit oscillates in its natural resonant mode. It is represented by the Greek letter omega ω. We have,

$$q = Q\cos(\omega t + \phi)$$

Taking second derivative of charge with respect to time we get,

$$\frac{di}{dt} = \frac{d^2q}{dt^2} = -\omega^2Q\cos(\omega t + \phi)$$

Substituting for q and $\frac{d^2q}{dt^2}$ in the differential equation $L\frac{d^2q}{dt^2} + \frac{1}{C}q = 0$ of the LC circuit, we obtain

$$-\omega^2Q\cos(\omega t + \phi) + \frac{1}{LC}Q\cos(\omega t + \phi) = 0$$

Canceling $Q\cos(\omega t + \phi)$ and rearranging lead to,

$$\omega^2 - \frac{1}{LC} = 0 \qquad \Rightarrow \omega = \frac{1}{\sqrt{LC}}$$

Electrical and Magnetic Energy Oscillations

In an LC oscillator circuit, electrical and magnetic energy continually oscillate back and forth between the electric field of the capacitor (C) and the magnetic field of the inductor (L). These oscillations result in the exchange of energy between the two energy storage elements in the circuit.

Electrical energy stored in the *LC* circuit at time t is,

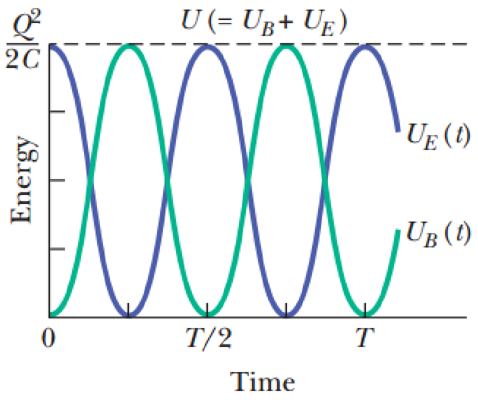
$$U_E = \frac{q^2}{2C} = \frac{Q^2}{2C}\cos^2(\omega t + \phi)$$

The magnetic energy is,

$$U_B = \frac{1}{2}i^2 L = \frac{1}{2}L\omega^2 Q^2 \sin^2(\omega t + \phi)$$

Substituting for $\omega = \frac{1}{\sqrt{LC}}$ gives us,

$$U_B = \frac{Q^2}{2C}\sin^2(\omega t + \phi)$$



Resonance:

Resonance is the phenomenon in which a system oscillates with **maximum amplitude** when it is driven at its natural frequency. In electrical circuits, resonance occurs when the ***impedance of the circuit*** reaches either a ***maximum*** (in parallel RLC circuits) or a ***minimum*** (in series RLC circuits) at a particular frequency.

The frequency at which resonance occurs is called the **resonant frequency**, and it depends only on the values of **inductance (L)** and **capacitance (C)** of the circuit. The resistance (R) affects the sharpness or damping of resonance but does not change the resonant

The resonance frequency of the LC circuit is given by,

$$f_r = \frac{\omega}{2\pi} = \frac{1}{2\pi\sqrt{LC}}$$

CHECKPOINT: A capacitor in an *LC* oscillator has a maximum potential difference of 17 V and a maximum energy of 160 μJ. When the capacitor has a potential difference of 5 V and an energy of 10 μJ, what are (a) the emf across the inductor and (b) the energy stored in the magnetic field?

ANSWER: $(\boldsymbol{a})5\ V;\ (\boldsymbol{b})150\ \mu J$

PROBLEM: A 1.5 μF capacitor is charged to 57 V by a battery, which is then removed. At time $t = 0$, a 12 mH coil is connected in series with the capacitor to form an LC oscillator.
(a) What is the potential difference $v_L(t)$ across the inductor as a function of time?
(b) What is the maximum rate $\left(\frac{di}{dt}\right)_{max}$ at which the current *i* changes in the circuit?

ANSWER: (a) At any time t during the oscillations, the loop rule and Fig. give us

$$v_L(t) = v_C(t)$$

that is, the potential difference v_L across the inductor must always be equal to the potential difference v_C across the capacitor, so that the net potential difference around the circuit is zero. Thus, we will find $v_L(t)$ if we can find $v_C(t)$, and we can find $v_C(t)$ from $q(t)$ with Eq. ($q = CV$). Because the potential difference $v_C(t)$ is maximum when the oscillations begin at time $t = 0$, the charge q on the capacitor must also be maximum then. Thus, phase constant ϕ must be zero; so

$$q = Q\cos\omega t$$

(Note that this cosine function does indeed yield maximum q $(= Q)$ when $t = 0$.) To get the potential difference $v_C(t)$, we divide both sides of Eq. by C to write

$$\frac{q}{C} = \frac{Q}{C}\cos\omega t \qquad \Rightarrow v_C = V_C\cos wt$$

Here, VC is the amplitude of the oscillations in the potential difference v_C across the capacitor. Next, substituting $v_C = v_L$, we get,

$$v_L = V_L \cos \omega t$$

We can evaluate the right side of this equation by first noting that the amplitude VC is equal to the initial (maximum) potential difference of 57 V across the capacitor. Then we find v with

$$\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{[(0.012\,H)(1.5 \times 10^{-6}\,F)]}} = 7454\frac{rad}{s} \approx 7500 rad/s$$

Thus,

$$v_L = (57\,V)\cos(7500\,rad/s)\,t$$

(b) Taking the derivative, we have

$$\frac{di}{dt} = \frac{d}{dt}(-\omega Q \sin \omega t) = -\omega^2 Q \cos \omega t$$

We can simplify this equation by substituting CV_C for Q (because we know C and V_C but not Q) and for ω.We get,

$$\frac{di}{dt} = -\frac{1}{LC} CV_C\ \cos \omega t = -\frac{V_C}{L} \cos \omega t$$

This tells us that the current changes at a varying (sinusoidal) rate, with its maximum rate of change being,

$$V_C = \frac{57\,V}{0.012\,H} = 4750\frac{A}{s} \approx 4800 A/s$$

PROBLEM 31-01: An oscillating LC circuit consists of a $75.0\,mH$ inductor and a $3.60\,\mu F$ capacitor. If the maximum charge on the capacitor is $2.90\,\mu C$, what are (a) the total energy in the circuit and (b) the maximum current?

PROBLEM 31-03: In a certain oscillating LC circuit, the total energy is converted from electrical energy in the capacitor to magnetic energy in the inductor in $1.50\,\mu s$. What are (a) the period of oscillation and (b) the frequency of oscillation? (c) How long after the magnetic energy is a maximum will it be a maximum again?

PROBLEM 31-04: What is the capacitance of an oscillating LC circuit if the maximum charge on the capacitor is $1.60\,\mu C$ and the total energy is $140\,\mu J$?

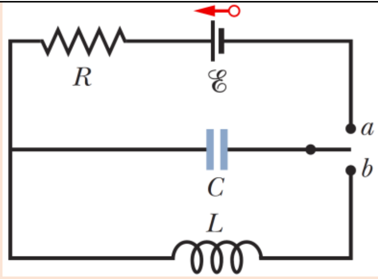
PROBLEM 31-05: In an oscillating LC circuit, $L = 1.10\,mH$ and $C = 4.00\,\mu F$. The maximum charge on the capacitor is $3.00\,\mu C$. Find the maximum current.

PROBLEM 31-06: A $0.50\,kg$ body oscillates in SHM on a spring that, when extended $2.0\,mm$ from its equilibrium position, has an $8.0\,N$ restoring force. What are (a) the angular frequency of oscillation, (b) the period of oscillation, and (c) the capacitance of an LC circuit with the same period if L is $5.0\,H$?

PROBLEM 31-09: In an oscillating LC circuit with $L = 50\,mH$ and $C = 4.0\,\mu F$, the current is initially a maximum. How long will it take before the capacitor is fully charged for the first time?

PROBLEM 31-10: LC oscillators have been used in circuits connected to loudspeakers to create some of the sounds of electronic music. What inductance must be used with a $6.7\ \mu F$ capacitor to produce a frequency of $10\ kHz$, which is near the middle of the audible range of frequencies?

PROBLEM 31-17: In Fig., $R = 14.0\ \Omega$, $C = 6.20\ \mu F$, and $L = 54.0\ mH$, and the ideal battery has emf $\varepsilon = 34.0\ V$. The switch is kept at a for a long time and then thrown to position b. What are the (a) frequency and (b) current amplitude of the resulting oscillations?

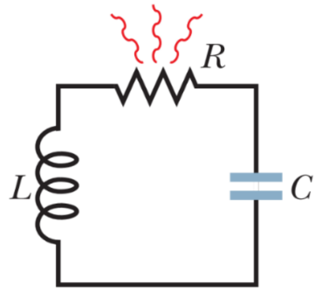


PROBLEM 31-21: In an oscillating LC circuit with $C = 64.0\ \mu F$, the current is given by $i = (1.60)\ sin(2500t + 0.680)$, where t is in seconds, i in amperes, and the phase constant in radians. (a) How soon after $t = 0$ will the current reach its maximum value? What are (b) the inductance L and (c) the total energy?

DAMPED OSCILLATIONS IN AN RLC CIRCUIT

An ***RLC circuit*** is an electrical circuit consisting of a resistor (R), an inductor (L), and a capacitor (C), connected in series or in parallel.

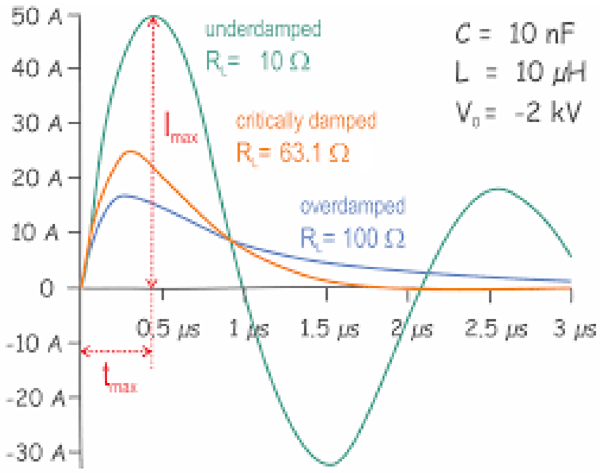
- This circuit exhibits damped oscillation behavior that gradually reduces the amplitude of the oscillations over time.
- The behavior of the circuit depends on the values of these components and the applied voltage or current.



When an ***RLC circuit*** is excited with an initial voltage or current, it exhibits three different types of damping: overdamping, underdamping, or critical damping.

<u>*Underdamping:*</u> In an underdamped RLC circuit, the damping is relatively low, allowing the current or voltage to oscillate before decaying.

- Underdamping occurs when the inductive and capacitive effects dominate over the resistance, leading to oscillations.
- The response is characterized by a decaying oscillation.

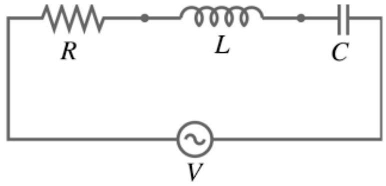
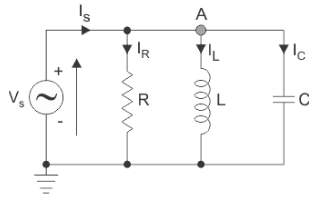


<u>*Critical damping:*</u> Critical damping represents the threshold between overdamping and underdamping.

- It occurs when the resistance is set to a value that causes the circuit to decay to zero as quickly as possible without oscillation.
- Critical damping results in the fastest decay of the oscillations without any overshoot or oscillatory behavior.

<u>*Overdamping:*</u> In an overdamped RLC circuit, the damping is relatively high, causing the current or voltage to decay without oscillation.

- Overdamping occurs when the resistance in the circuit is much larger than the other components, dominating the system's behavior.
- The response is characterized by a slow decay of the amplitude without any oscillatory behavior.

In Series RLC Circuit Configuration	In Parallel RLC Circuit Configuration
	
– R, L, and C are connected one after another in a single path. – ***At resonance***, impedance is ***minimum***, and current is ***maximum***. – Used in ***tuned circuits***, like radios & filters. – Total impedance: $Z = R + j\left(\omega L - \frac{1}{\omega C}\right)$ – Resonance Condition: $\omega L = \dfrac{1}{\omega C} \qquad \Rightarrow f_r = \dfrac{1}{2\pi\sqrt{LC}}$	– R, L, and C are connected in parallel across the same two nodes. – At resonance, impedance is ***maximum***, and current is ***minimum***. – Often used in **oscillator** and **tank circuits**. – Total impedance: $Y = \frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right)$ – Resonance Condition: $\omega C = \dfrac{1}{\omega L} \qquad \Rightarrow f_r = \dfrac{1}{2\pi\sqrt{LC}}$

Analysis of RLC Circuit:

With a resistance R present, the total *electromagnetic energy* U of the circuit (the sum of the electrical energy and magnetic energy) is no longer constant; instead, it decreases with time as energy is transferred to thermal energy in the resistance. Because of this loss of energy, the oscillations of charge, current, and potential difference continuously decrease in amplitude, and the oscillations are said to be *damped*.

An equation for the total electromagnetic energy U in the circuit at any instant is the sum of the stored electric and magnetic energy (since, the resistance does not store electromagnetic energy), we get,

$$U = U_B + U_E = \frac{1}{2}i^2L + \frac{q^2}{2C} \dots\dots\dots (1)$$

However, the total energy decreases over time; as the energy is transferred to thermal energy. The rate at which energy is transferred to thermal energy ($P = i^2R$) due to flow of current through the resistor is given by,

$$\frac{dU}{dt} = -i^2R \dots\dots\dots (2)$$

where the minus sign indicates that U decreases. By differentiating Eq. (1) with respect to time and then substituting the result in Eq. (2), we obtain

$$\frac{dU}{dt} = iL\frac{di}{dt} + \frac{q}{C}\frac{dq}{dt} = -i^2R \dots\dots\dots (3)$$

Substituting $i = \frac{dq}{dt}$ and $\frac{di}{dt} = \frac{d^2q}{dt^2}$, we obtain

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q = 0 \dots\dots\dots (4)$$

which is the differential equation for damped oscillations in an *RLC* circuit.

Charge Decay: The solution to Eq. (4) is,

$$q = Qe^{-Rt/2L}\cos(\omega' t + \phi) \;\dots\dots\dots (5)$$

In which,

$$\omega' = \sqrt{\omega^2 - \left(\frac{R}{2L}\right)^2} = \omega\sqrt{1 - (LC)\left(\frac{R}{2L}\right)^2} = \omega\sqrt{1 - \left(\frac{R}{2}\sqrt{\frac{C}{L}}\right)^2} = \omega\sqrt{1-\zeta^2} \;\dots\dots\dots (6)$$

Here $\omega = \frac{1}{\sqrt{LC}}$, as with an undamped oscillator, and $\zeta\ (zeta)$ is equal to $\zeta = \frac{R}{2}\sqrt{\frac{C}{L}}$.

Equation (5) describes a sinusoidal oscillation (the cosine function) with an exponentially decaying amplitude of charge on the capacitor, $Qe^{-Rt/2L}$, in a damped RLC circuit. The angular frequency ω' of the damped oscillations is always less than the angular frequency ω of the undamped oscillations.

Energy Decay: Total electromagnetic energy U of the RLC circuit as a function of time can be obtained by monitoring the energy of the electric field in the capacitor ($U_E = \frac{q^2}{2C}$). Thus, by substituting Eq. (5) we get,

$$U_E = \frac{q^2}{2C} = \frac{\left[Q\, e^{-\frac{Rt}{2L}}\cos(\omega' t+\phi)\right]^2}{2C} = \frac{Q^2}{2C}e^{-\frac{Rt}{L}}\cos^2(\omega' t + \phi) = \frac{Q^2}{2C}e^{-\frac{t}{\tau}}\cos^2(\omega' t + \phi) \;\dots\dots\dots (7)$$

Thus, the energy of the electric field oscillates according to a cosine-squared term, and the amplitude of that oscillation decreases exponentially with time.

Underdamping	Critical damping	Overdamping
$\zeta < 1$ Thus, $R < 2\sqrt{\frac{L}{C}}$	$\zeta = 1$ Thus, $R = 2\sqrt{\frac{L}{C}}$	$\zeta > 1$ Thus, $R > 2\sqrt{\frac{L}{C}}$

Impedance:

Impedance of resistor is $X_R = R$.

Impedance of inductor is $X_L = \omega L = 2\pi f L$

Impedance of capacitor is $X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$

Resonance: In a standard RLC Circuit, the resonant frequency is independent of resistance (R). It depends only on the values of inductance (L) and capacitance (C).

The resistance (R) does not affect the resonant frequency, but it controls the damping of oscillations and the sharpness of resonance (i.e., the bandwidth or quality factor of the circuit).

Time constant: The time constant of an RLC circuit, denoted by τ (tau), that determines the decay rate of the oscillations. It is given by the expression $\tau = L/R$, where L is the inductance and R is the resistance in the circuit.

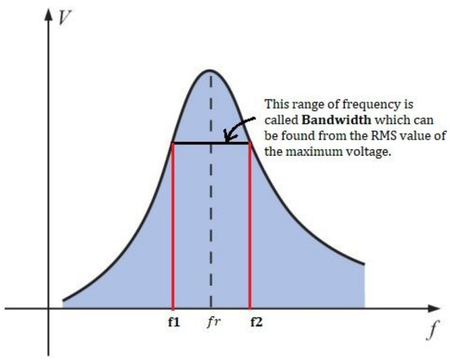
Quality Factor: The Q-factor, or Quality Factor, is a dimensionless parameter that describes how efficiently a resonant system stores energy compared to how quickly it loses energy.

- It indicates the ***sharpness of resonance*** and the ***degree of damping*** in oscillatory systems.
- The **Quality Factor**, or Q**-Factor**, describes how underdamped an oscillator or resonator is, and characterizes a resonant system's bandwidth relative to its center (resonant) frequency.
- For a parallel RLC circuit, assuming L and C are ideal, the Q-factor is theoretically given by,

$$Q = R\sqrt{\frac{C}{L}}$$

In an **RLC circuit**, the **quality factor** (or **Q-factor**) is defined as the ratio of resonance frequency to the difference of its neighboring frequencies so that their corresponding signal is $\frac{1}{\sqrt{2}}$ times of the peak value.

- The **quality factor** (or **Q-factor**) is a measure of how "sharp" or "selective" the resonance is.
- For instance, when Q is large, the peak of the graph is sharp and the bandwidth is small.

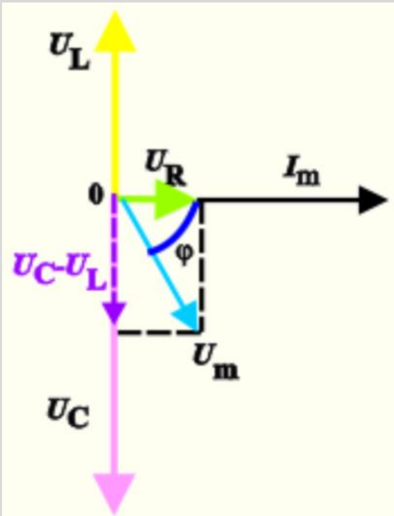


$$Q = \frac{f_r}{\Delta f} \qquad and \qquad \Delta f = f_2 - f_1$$

Here, f_r is the resonant frequency, and Δf the bandwidth, where the bandwidth is the difference between two frequencies f_1 and f_2 on either side of f_r at which the ***output signal drops to*** $\boldsymbol{\frac{1}{\sqrt{2}}}$ ***times*** (or approximately 70.7%) ***of the peak value*** $\boldsymbol{f_r}$.

<u>NOTE:</u> A phasor is an "arrow" that we use to plot the current and voltage values on individual components of the circuit into a phasor diagram. Its magnitude reflects the amplitude of the voltage or current, and its direction indicates the phase angle.

<u>Drawing a phasor diagram for a series circuit:</u> We plot the values of voltage and current on individual components in the AC circuit into the phasor diagram. For an RLC circuit and the given quantities the phasor diagram looks like the one presented in figure.



- The current is of the same size on all the components, the phasor of current I_m is therefore the same for all the components and is usually drawn in the positive direction of the *x*-axis.
- The phasor of voltage is on the resistor U_R parallel to the current phasor, because the phase difference between the voltage and current is zero – in this case voltage and current are in phase. In the figure the phasor is illustrated by green.
- The voltage on the coil U_L leads the current by π/2 (quarter of a period), therefore we draw its phasor pointing upwards – in the positive direction of the imaginary *y*-axis. We consider the fact that the phasors rotate in a counter clockwise direction. In the figure, this phasor is represented by yellow.

- The current on the capacitor leads the voltage U_C by π/2. Therefore we draw the phasor pointing downward – that is in negative direction of the y-axis. This phasor is represented by pink.
- The amplitude of the overall voltage is obtained by a "vector sum" of phasors of the voltage on individual components. First, we subtract the voltage on the capacitor U_C from the voltage on the coil U_L (in the picture drawn in purple). Then we add this vector and the vector of the voltage on the resistor U_R. The phasor of the voltage amplitude of the entire circuit is represented by light blue.
- A phase difference between the voltage and the current is said to be the angle φ between the current phasor and the overall voltage phasor. The angle φ is drawn by navy blue.

PROBLEM: A series RLC circuit has inductance $L = 12\, mH$, capacitance $C = 1.6\, \mu F$, and resistance $R = 1.5\, \Omega$ and begins to oscillate at time $t = 0$.

(a) At what time t will the amplitude of the charge oscillations in the circuit be 50% of its initial value? (Note that we do not know that initial value.)

(b) How many oscillations are completed within this time?

ANSWER: (a) When the charge amplitude has decreased to $0.50Q$ *we can write,*

$$Q\, e^{-\frac{Rt}{2L}} = 0.50\, Q$$

Taking the natural logarithms of both sides (to eliminate the exponential function), we have

$$-\frac{Rt}{2L} = \ln 0.50$$

Solving for the time t and then substituting given data yield

$$t = -\frac{2L}{R}\ln 0.50 = -\frac{2(12 \times 10^{-3}\, H)(\ln 0.50)}{(1.5\, \Omega)} = 0.0111\, s \approx 11\, ms$$

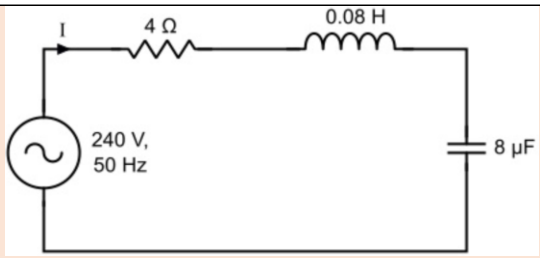
(b) In the time interval $\Delta t = 0.0111\, s$, the number of complete oscillations is

$$\frac{\Delta t}{T} = \frac{\Delta t}{2\pi\sqrt{LC}} = -\frac{0.0111}{2\pi[(12 \times 10^{-3}\, H)(1.6 \times 10^{-6}\, F)]^{\frac{1}{2}}} \approx 13$$

PROBLEM 31-24: A single-loop circuit consists of a $7.20\, \Omega$ resistor, a $12.0\, H$ inductor, and a $3.20\, \mu F$ capacitor. Initially the capacitor has a charge of $6.20\, \mu C$ and the current is zero. Calculate the charge on the capacitor N complete cycles later for (a) $N = 5$, (b) $N = 10$, and (c) $N = 100$.

PROBLEM 31-26: In an oscillating series RLC circuit, find the time required for the maximum energy present in the capacitor during an oscillation to fall to half its initial value. Assume $q = Q$ at $t = 0$.

PROBLEM: A $240\, V$, $50\, Hz\, AC$ supply is applied to a coil of $0.08\, H$ inductance and $4\, \Omega$ resistance connected in series with a capacitor of $8\, \mu F$. Calculate the Impedance, Circuit current, Phase angle between voltage and current, and power consumed.



ANSWER: Here,

$$X_L = \omega L = 2\pi f L = 2\pi(50)(0.08) = 25.12\Omega$$

And

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f L} = \frac{1}{2\pi(50)(8 \times 10^{-6})} = 398.09\Omega$$

Thus, the impedance of the circuit

$$Z = \sqrt{(R)^2 + (X_L - X_C)^2} = \sqrt{(4)^2 + (25.12 - 398.09)^2} = 372.99\ \Omega$$

Circuit current, $i = \frac{V}{Z} = \frac{240}{372.99} = 0.643\ A$

Phase angle between voltage and current

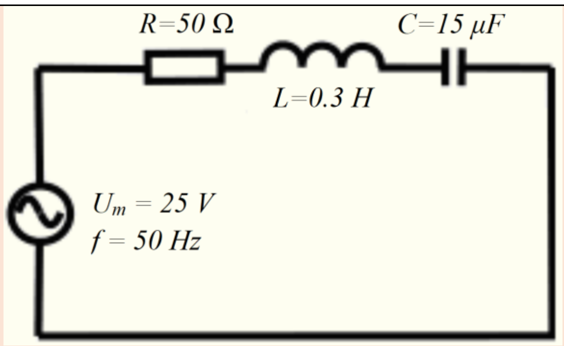
$$\Phi = \tan^{-1}\left(\frac{X_L - X_C}{R}\right) = \tan^{-1}\left(\frac{25.12 - 398.09}{4}\right) = -89.38°$$

The negative sing of phase angle shows that current is leading the voltage.

Power Factor, $\cos\phi = \frac{R}{Z} = \frac{4}{372.99} = 0.01072$

Therefore, Power consumed $P = VI\cos\Phi = (240)(0.643)(0.01072) = 1.654\ W$

PROBLEM: An AC circuit is composed of a serial connection of a resistor with resistance $50\ \Omega$, a coil with inductance $0.3\ H$ and a capacitor with capacitance $15\ \mu F$. The circuit is connected to an AC voltage source with amplitude $25\ V$ and frequency $50\ Hz$. Determine the amplitude of electric current in the circuit and the phase difference between the voltage and the current.



ANSWER: Given that, Resistance of resistor, $R = 50\ \Omega$

Inductance of the coil, $L = 0.3$ H

Capacitance, $C = 15\ \mu$F $=15\cdot10^{-6}$ F

Amplitude of AC voltage source, $Um = 25$ V

Frequency of source, $f = 50\ Hz$

Resistor, coil and capacitor are connected in series.

From Ohm's law we know that:

$U_C = I_m X_C$, $U_L = I_m X_L$, $U_R = I_m R$.

Since the current through all the components is the same, the impedances of individual elements are proportional to the voltage, so we can draw a diagram similar to the voltage phasors.

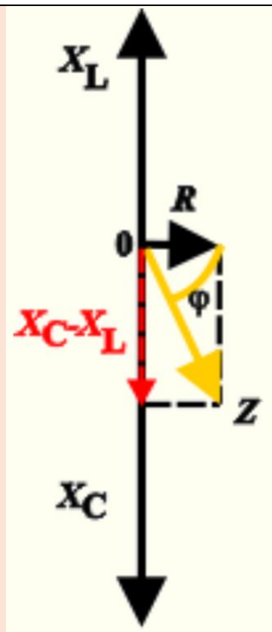
According to the rectangular triangle we can see in the phasor diagram. The impedance Z is evaluated by using Pythagorean theorem.

$$Z^2 = R^2 + (X_C - X_L)^2$$
$$\Rightarrow Z^2 = R^2 + (X_L - X_C)^2$$

The difference between the relationships is whether the current leads voltage or the voltage leads the current. The size of the impedance Z is however not affected

By substituting the relations of inductance and capacitance we obtain:

$$Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$



The formula for expressing the impedances Z from Ohm's law is,

$$Z = \frac{U_m}{I_m} = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} \qquad \Rightarrow I_m = \frac{U_m}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

$$I_m = \frac{25\ V}{\sqrt{(50\ \Omega)^2 + \left(2\pi(50\ Hz)(0.3H) - \frac{1}{2\pi(50\ Hz)(15\times 10^{-6}\ F)}\right)^2}} = 0.2\ A$$

The phase shift is expressed from the phasor diagram usually in the form,

$$\tan\varphi = \frac{(U_L - U_C)}{U_R} = \frac{I_m\omega L - \frac{I_m}{\omega C}}{I_m R} = \frac{\omega L - \frac{1}{\omega C}}{R} = \frac{X_L - X_C}{R}$$

When we draw a phasor diagram and a phase difference, the formula

$$\tan\varphi = \frac{(U_L - U_C)}{U_R} = \frac{X_L - X_C}{R} = \frac{2\pi(50\ Hz)(0.3\ H) - \frac{1}{2\pi(50\ Hz)(15\times 10^{-6}\ F)}}{50\ \Omega} = -2.4$$

Therefore, $\varphi = \tan^{-1}(-2.4) = -67°$

A negative value says that the current leads the voltage by about 67° (since the current leads the voltage, this circuit acts as a capacitor).